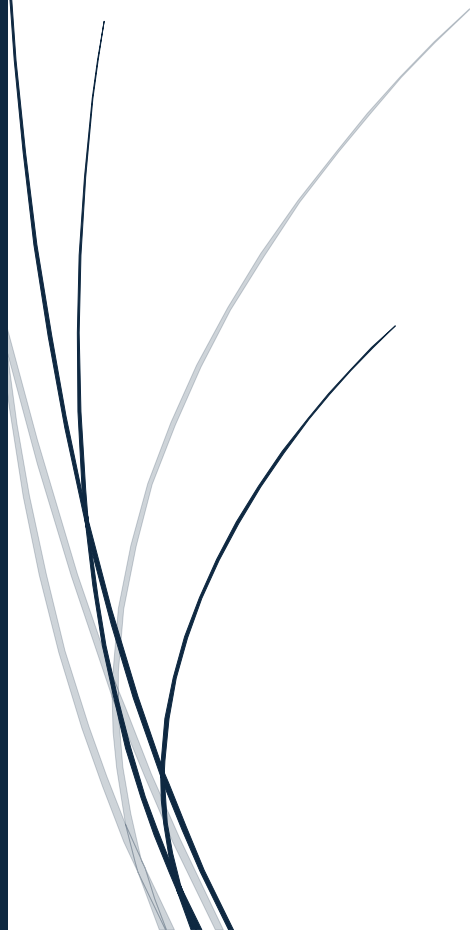


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Student Access Card App
Design Document
CMPU3026 TU856/3



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Introduction

This project creates a digital student access card app designed to enable TUD students to access campus facilities (library, printer) using a student ID card interface. The app integrates facial recognition, ID/password login, and uses TUD's theme colors (Blue, White, Green) to provide a familiar user experience. Primary features include digital ID presentation, library access, printing access.

Objectives

- Develop a digital student ID card for TUD students with login options (Facial ID, Student ID/Password, Passkey).
- Include core functionalities: library access, printer usage, credit, and notifications.
- Provide a user-friendly interface with accessible design, high contrast for visibility, and support for Dark Mode.

App Design

Colour Scheme and Layout

Colors:

- TUD theme colors: Blue, White, Green
- Backgrounds: White (Light Mode) and Black (Dark Mode) for contrast and accessibility.

User Interface Design:

- Header Section: Displays the student's name and ID number, profile picture in a framed layout, and additional details like course and faculty.
- Digital Card Section: Offers a visual representation of the student card with swipe functionality to reveal additional information.
- Action Cards Section: Quick-access icons for Library, Printer, and Credit Management.

- Bottom Navigation: Home, Profile, Library, and Printing navigation options.

User Information Display

Header Section:

- Profile Picture: Clear, high-resolution image to a physical ID photo.
- Student Details: Displays student name, ID number, course details, academic year, and faculty (e.g., Computer Science, 3rd year).
- Graduation/Expiry Date: Sets graduation year as the ID expiration.

Digital Card Display:

- Expandable Section: Displays additional card details like barcode (for scanner compatibility in campus/library/printer).
- Swipe Interaction: Smooth expand/collapse gesture to reveal/hide card details.

Action Cards

Provides quick access to essential services:

1. Library Access Icon: Book icon for easy recognition.
2. Printer Access Icon: Printer symbol for instant access.
3. Credits/Top-Up: Euro sign for managing printer credits.

Notifications

Real-time notifications include:

- Library alerts (e.g., available books or return due dates).
- Printer alerts (e.g., “Document ready for printing” notification).
- Icon badges and subtle alert sounds for attention.

Key Functionalities and Requirements

1. Multi-Activity Design: Three or more activities, excluding login, where data is shared using intents.
2. Database Integration : Stores student details, usage history.
3. Sensor Usage: Uses biometric (Facial ID) and sensors for login security.
4. Custom Views & Gestures: Expandable card view and swipe gesture for additional card details.
5. GitHub Version Control: Consistent commits to a shared GitHub repository.
6. Dark Mode & Accessibility: Support for Dark Mode and high-contrast UI.
7. State Management & Optimization: Efficient memory usage with optimized loading speeds.
8. Notification Integration: Push notifications for library and printing updates.

User Interaction Design and UX

The design follows Android UX guidelines for intuitive navigation:

- Login Screen: Allows entry via Facial ID, Passkey, or Student ID/Password. Accessibility features include keyboard shortcuts and auto-focus.
- Home Screen: Shows an overview of the digital student card, access icons, and profile information.
- Notification System: Alerts shown through notification tray with relevant icons (e.g., library book icon for library notifications).

Technical Implementation

- Data Handling: database for storing student data and transaction history.
- Sensors and Biometrics: Utilizes facial recognition for secure access.

- Custom Views & Swipe Gestures: Utilizes Android's View animations for a smooth, expandable card interface.
- Continuous Git Integration: Update and commit regularly with documented changes.

Accessibility and User-Centered Design

The app is designed with accessibility at the forefront:

- Contrast Settings: Ensures visibility in both Light and Dark Modes.
- Font Scaling: Supports Android's font scaling for readability.
- Icons and Labels: Clearly labelled icons for navigation with tooltips and haptic feedback.

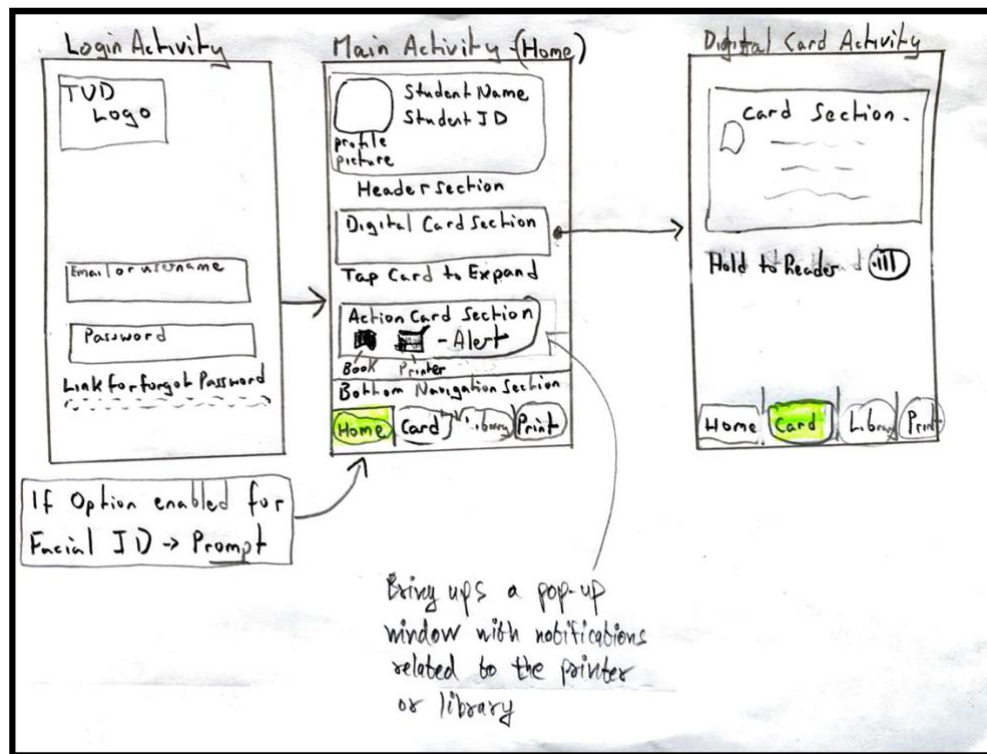
Future Enhancements

Potential improvements:

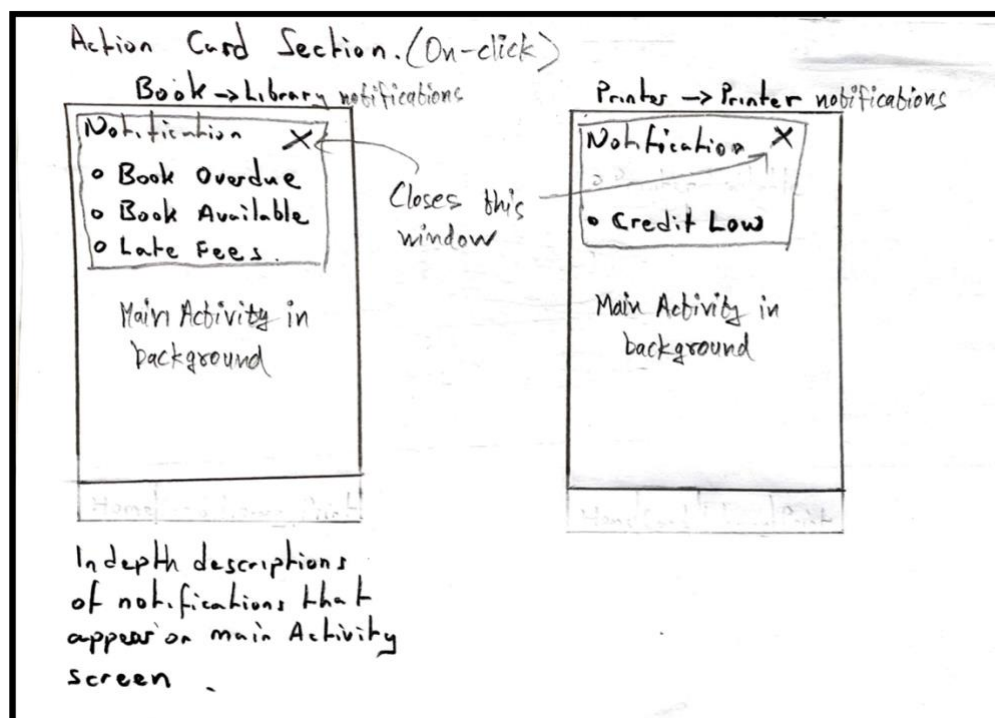
- NFC Integration: To facilitate contactless ID access for future iterations.
- Advanced Notification Settings: Customized alerts for library or printer.

Wireframes

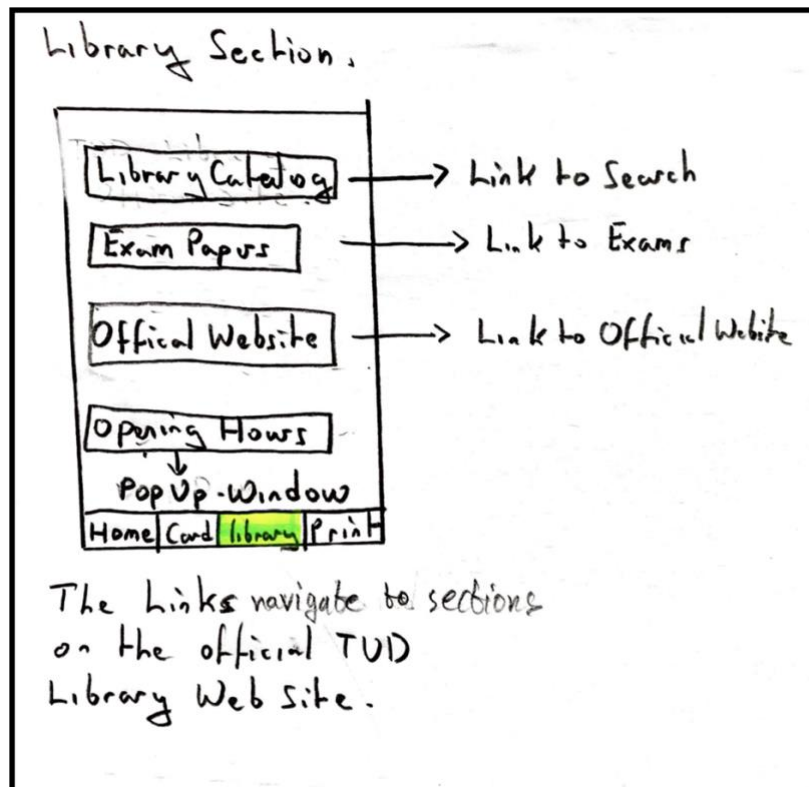
Home Screen, Log in and Digital Card Screen



Notification Pop-Up windows



Library



Printer

